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DSC-HUB

Grow Environment Controller — User Manual

Firmware: rev 3 (dsc-hub.yaml, ~1,631 lines, ESPHome 2026.6.x) · **Device:** `airquality` / DSC-HUB (ESP32 WROOM-32) · **Document date:** 7 July 2026

Scope: Two-tent grow room (4×8 + 2×4), software-defined damper ventilation, five-rung appliance escalation ladder, device-local photoperiod, OLED dial UI, Home Assistant integration.

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1 · System Overview

DSC-HUB is a standalone ESP32 environmental controller for a two-tent cannabis grow (one 4×8 flowering tent, one 2×4 tent) inside a dedicated room. It manages ventilation, lighting schedule, and climate escalation locally on the device, with Home Assistant acting as a remote control surface and the switching layer for mains appliances.

The design philosophy, in order:

1. **Fans first.** Airflow is cheap, fast, and quiet. The controller always tries to solve a climate problem with fan blending before waking any appliance.
2. **Appliances on demonstrated failure.** Dehumidifier, humidifier, grow mat, heater and AC are only demanded after a condition has *persisted despite* the fans (the escalation ladder, §6).
3. **Survive infrastructure loss.** The photoperiod runs on-device via SNTP. If Home Assistant, WiFi, or the network dies, the lights keep their schedule and the fans keep their curve. HA is convenience, not a dependency.
4. **Fail toward ventilation, never toward heat.** Every failure path (sensor fault, power loss, hub offline) lands on moving air and de-energised heat appliances.

Where the truth lives

The firmware source is `dsc-hub.yaml`. Its header comment block is the canonical short-form spec; this manual is the long-form explanation. If the two ever disagree, the YAML that is actually flashed wins — update this document.

2 · Hardware

Component	Detail	Role
Controller	ESP32 WROOM-32 dev board	Runs ESPHome firmware; all logic is local
Display	SH1106 128×64 OLED, I ² C @ 400 kHz (GPIO21 SDA / GPIO22 SCL)	Status pages, menu UI, screensaver
Input	Rotary encoder with push, + BAK button (GPIO14), CON button (GPIO33)	All local navigation and adjustment
Climate sensor	DHT22 (temperature + RH)	Feeds VPD calculation, curves, ladder, watchdog
CO ₂	Analog CO ₂ estimate on GPIO34 (ADC)	Informational (CO ₂ e on System Info page)
Fan 1	6" inline — Exhaust (Outside) , carbon filter #1	Dumps heat/moisture out of the building
Fan 2	6" inline — Exhaust (Room) , carbon filter #2	Recirculates tent air into the room lung
Fan 3	4" inline — Intake (Main)	Fresh air, room → 4×8; slaved to total exhaust
Fan 4	4" inline — Intake (2×4)	Fresh air, room → 2×4 (cascades into 4×8)
Light	SF1000, PWM dimmable	Photoperiod with sunrise/sunset ramping
Ladder appliances	Dehumidifier, humidifier, grow mat, 750 W fan heater, portable AC (cool @ 20 °C)	Switched by HA smart plugs following DSC-HUB demand signals

GPIO protection — mandatory on the PWM lines

A previous WROOM-32 was killed by gate-drive overvoltage from a PWM driver module back-feeding a GPIO (suspected CMOS latch-up + regulator failure). Any board driving external PWM modules must have a **330 Ω–1 kΩ series resistor plus a 3.3 V zener/Schottky clamp** on each PWM signal line — or an opto-isolated gate driver. Do not connect a bare GPIO to a fan/light driver module. See §12.

History note (rev 2 hardware change)

The original 250 W LED array was removed and its channel (GPIO19) repurposed to drive the fourth 6" inline fan, enabling the two-exhaust damper architecture in §4. Any dashboard or automation still referencing the 250 W light is stale.

3 · Room & Tent Layout — Ducting, Intakes, Exhausts, Appliance Placement

3.1 Top-down room map

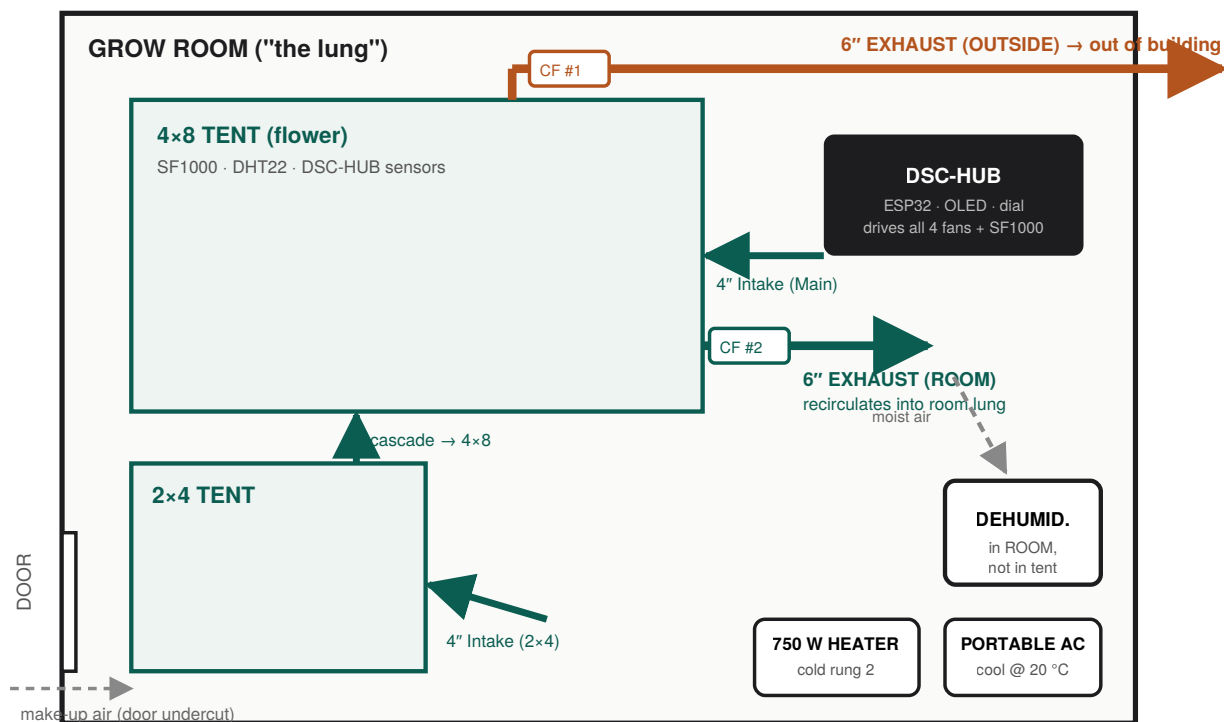


Fig. 1 — Room map. Green = fresh/recirculated air paths (managed by DSC-HUB PWM). Orange = the only path that leaves the building. Grey dashed = passive/appliance flows. CF = carbon filter.

3.2 Ducting & flow schematic

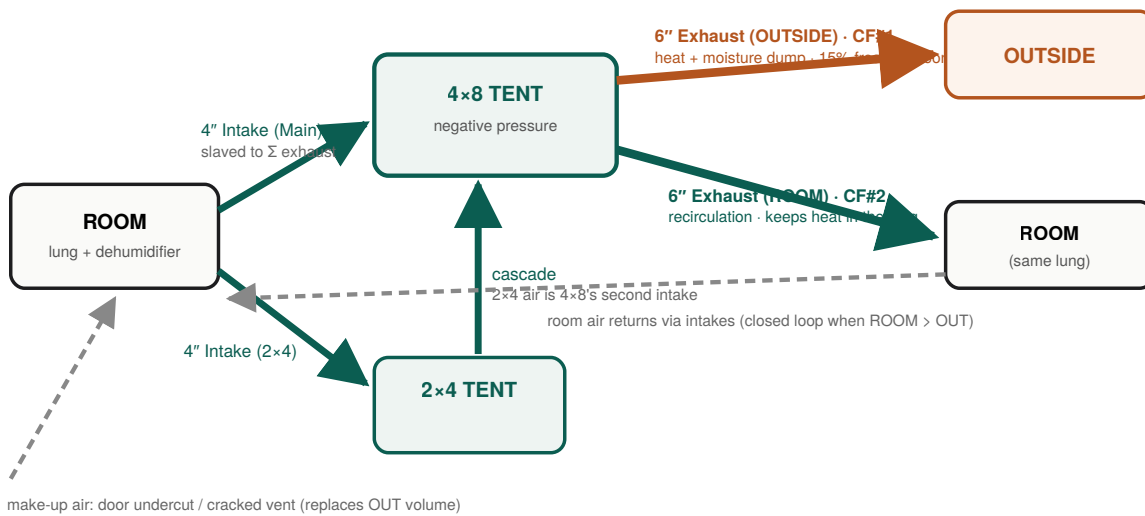


Fig. 2 — Flow topology. The blend ratio between the two 6" exhausts is the damper: OUT-heavy = purge mode, ROOM-heavy = closed-loop recirculation through the dehumidified lung.

3.3 Appliance placement & reasoning

Appliance	Placement	Why
Dehumidifier	In the room, near the ROOM exhaust discharge — not inside a tent	Three reasons. (1) A dehumidifier is a heat source; inside a sealed tent it fights the temperature target it's supposed to protect. (2) The room is the lung: the ROOM exhaust delivers moist tent air to it, the dehumidifier strips the moisture, and the intakes return dry air — one machine serves both tents. (3) Floor space and drainage — the tank/hose lives outside the canopy.
Humidifier	Room, near the intake pickups	Same lung logic in reverse: humidified room air is drawn into both tents by the intakes. Never point a humidifier at the DHT22 or you'll create a false RH reading and oscillation.
750 W heater	Room, clear of tent fabric, never inside a tent	Heats the intake supply air rather than a single tent; the intakes then distribute it. Fire safety: fan heaters and tent fabric don't share a volume. Fail-safe: HA cuts the plug if the hub goes unavailable >30 s.
Grow mat	Under pots (root zone), 2x4 primarily	Cold rung 1: root-zone temperature matters more than air temperature for early stages, and a mat is ~50 W vs 750 W — the ladder tries it first.
Portable AC	Room, set to cool @ 20 °C, ducted per its own exhaust requirements	Cools the lung/supply air. It is the last rung for heat because it's the loudest, thirstiest appliance — the OUT exhaust gets 5 minutes at full tilt first.
Carbon filters	CF#1 on the OUTSIDE exhaust, CF#2 on the ROOM exhaust	Both exhaust paths are scrubbed, so odour is controlled whether the damper is purging or recirculating. Recirculation without CF#2 would perfume the house.

Ducting practice

Keep runs short and straight; every 90° bend costs roughly a duct-diameter of effective flow. The OUT run is the longest — it earns the 6" diameter. Seal tent duct ports around the duct, not just over it: the negative-pressure math in §4 assumes leaks are small relative to intake area. The 2×4→4×8 cascade duct should be as short as the tent spacing allows.

4 · The Software-Defined Damper — Design Reasoning

A conventional single-exhaust tent has one lever: exhaust speed. Every problem — too hot, too humid, too dry, low CO₂ — gets the same answer, and in a Newcastle winter that answer is wrong half the time: purging tent air outside dumps the light's heat you just paid for, and pulls in cold, damp coastal air.

DSC-HUB replaces the single lever with a **blend ratio between two 6" exhausts**:

Fan	Destination	Ramps on	Character
Exhaust (Outside) — CF#1	Out of the building	Temperature over target; RH overflow when the dehumidifier can't cope; plus a permanent 15% fresh-air floor	The purge. Loses heat, gains fresh CO ₂ . Summer / lights-on workhorse.
Exhaust (Room) — CF#2	Back into the room lung	VPD / RH demand	The recirculator. Moves moisture to the dehumidifier <i>without</i> losing heat. Winter / cold-but-humid workhorse.

Blending the two percentages *is* the damper — there is no mechanical flap, no servo, nothing to jam. The controller separates the two problems that a single fan conflates:

- **Heat is an OUT problem.** Only exhausting outside removes energy from the system.
- **Moisture is a ROOM problem.** Recirculating through the dehumidified lung removes water while conserving heat — exactly what a cold, humid NSW coastal winter demands.

4.1 Negative pressure & the intake slaving rule

The 4" Intake (Main) is **slaved to total exhaust** (OUT + ROOM) and always commanded slightly below it. Exhausting more than you intake keeps both tents at slight negative pressure, which:

- guarantees all air leaves through a carbon filter (odour control),
- keeps tent walls pulled inward (visual leak indicator), and
- makes the 2×4 cascade deterministic — the 4×8's depression pulls the 2×4's air through the cascade duct, so the 2×4 gets flow-through without its own exhaust fan.

4.2 Why the 15% fresh-air floor

In a fully-recirculating (ROOM-dominant) state the loop is closed: plants consume CO₂ and nothing replaces it. The OUT exhaust therefore never drops below 15%, which continuously bleeds a little air outside and drags fresh make-up air in through the door undercut. CO₂ never fully depletes, at the cost of a small, predictable heat leak the escalation ladder can compensate for.

4.3 Emergency posture

Above 35 °C the failsafe forces **OUT 100% / ROOM 0% / both intakes 100%** — a once-through purge to outside. Recirculation during a heat emergency would launder hot air back into the supply; the failsafe deliberately breaks the loop.

5 · Control Model

5.1 The four layers

1. **Grow Stage** (select) — picking a stage auto-loads the VPD / RH / temperature targets and photoperiod length for that stage. Fine-tuning the number sliders afterwards persists across reboots; only a stage *change* resets them.
2. **Control Strategy** (select) — which variable the fan curve chases: **VPD** (grower-grade, default), **Temperature**, or **Humidity**.
3. **Tent Full Auto Mode** (switch) — arms the two-axis fan auto-curve (§4).
4. **Auto Photoperiod** (switch) — the *device* runs the light schedule locally via SNTP; HA only sets parameters (stage → duration, Lights-On Time → start).

5.2 Grow stages

Ten stages, Germination → Dry Mode, each carrying researched VPD / RH / temperature targets validated against 2026 horticultural consensus. Headline VPD bands:

Phase	VPD target band	Notes
Seedling / clone	0.4 – 0.8 kPa	High RH, gentle air; grow mat is the preferred heat rung
Vegetative	0.8 – 1.2 kPa	
Early flower	1.0 – 1.3 kPa	
Late flower	1.2 – 1.6 kPa	Final 2 weeks: 20–23 °C lights-on / 18–20 °C off, RH trending toward ~30% for resin and rot protection
Dry Mode	—	~19 °C / ~60% RH / dark / low fans. Drying wants <i>low, indirect</i> airflow over 7–14 days — the ~20% exhaust baseline covers it; don't blast it.

The full per-stage table (all ten stages, exact numbers) lives in the `apply_stage` section of `dsc-hub.yaml` — that is the source of truth.

Newcastle seasonal note

Coastal NSW's binding constraints are **heat in summer** and **humidity in flower** (mould pressure). Standard mitigation: run the photoperiod overnight in summer so lights-on coincides with cool ambient — this is exactly why Lights-On Time is an exposed HH:MM setting (default 06:00; an evening start is the summer play). The dehumidifier rung exists primarily to save flower runs from bud rot.

5.3 Photoperiod & sunrise/sunset ramping

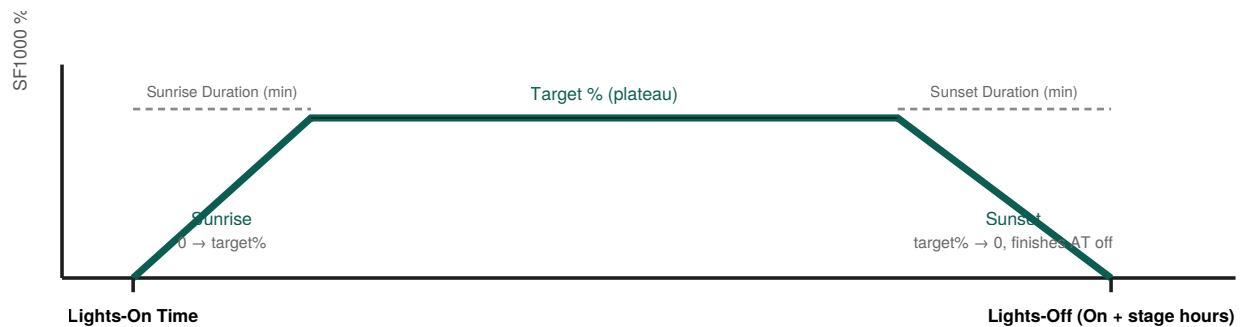


Fig. 3 — Both ramps live *INSIDE* the photoperiod window and auto-scale if their durations would overlap. The schedule runs on-device (SNTP, timezone Australia/Sydney; HA time as backup) and survives HA/network loss.

5.4 Manual semantics — the split override

Control	Scope	Behaviour
Tent Manual Override	Fans only	Device stops driving fans (no curve, no slaving/roles); your fan values stick. The photoperiod keeps driving the light. Mutually exclusive with Full Auto. The CON button toggles this — a deliberate panic hold.
Manual Light Hold	Light only	Engages automatically the moment you hand-adjust the SF1000 (dial or HA). The schedule keeps ticking but stops touching the light. Self-heals at the next lights-off boundary (a daytime tweak resumes overnight — no herm trap). Also clears when Manual Override turns off, when Auto Photoperiod toggles on, and on reboot (deliberately not persisted — after a power cycle the schedule is authoritative).

5.5 Priority order

1 · EMERGENCY FAILSAFE (>35 °C) OUT 100% · ROOM 0% · intakes 100% (once-through purge) · light cut · heat demands off, AC on
2 · SENSOR-FAULT SAFE MODE (watchdog) no valid temp for 3 min → safe ventilation, ALL appliance demands off (climate unknown)
3 · MANUAL (fans) / MANUAL LIGHT HOLD your values stick; safety layers above still win
4 · AUTO PHOTOPERIOD + FULL AUTO CURVE normal operation: two-axis damper blend + device-local light schedule

Higher row always wins. Escalation-ladder timers reset whenever rows 1–2 engage, so an emergency never counts as "fans failed".

Fig. 4 — Priority stack, highest wins.

6 · Appliance Escalation Ladder

Demands are **soft signals** — template switches HA automations watch to switch the real smart plugs. The device never carries mains current. Each appliance has an **Auto** arm switch (enables its rung) and a **Demand** switch (the signal). Demands only engage when the condition has *persisted despite the fans*; timers reset the instant the condition clears, and whenever the failsafe or watchdog engages.

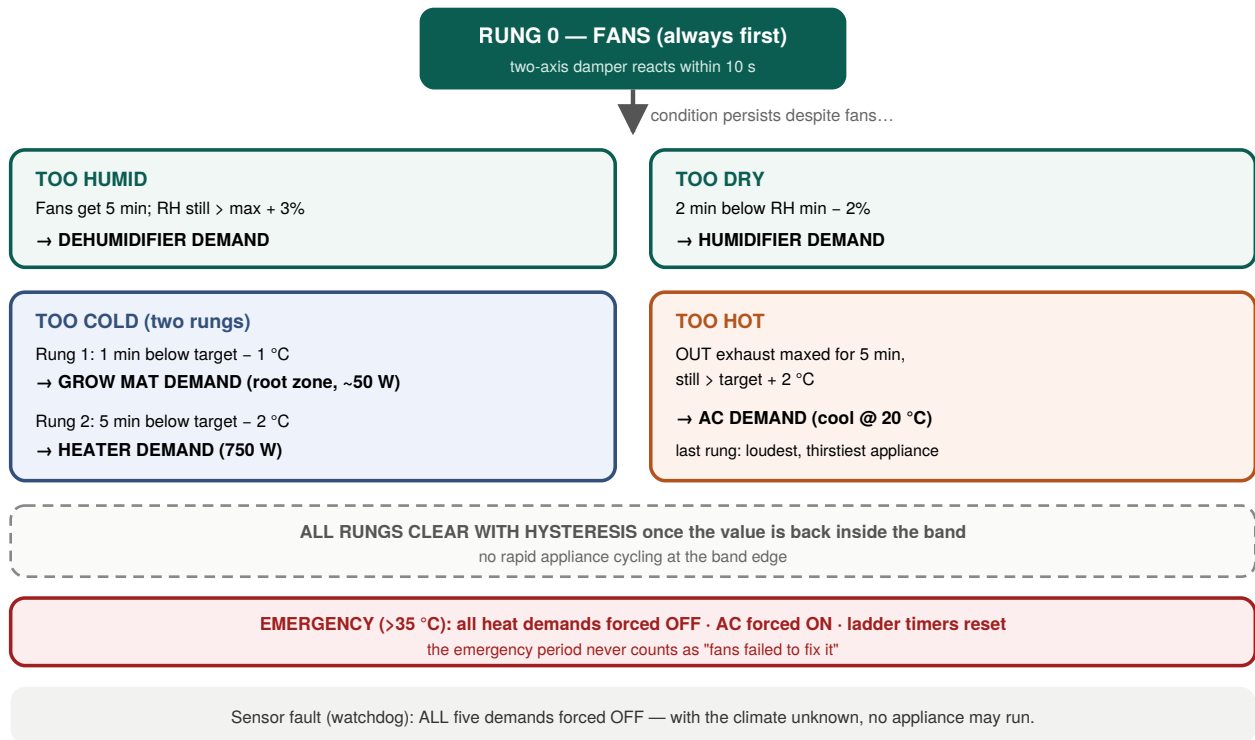


Fig. 5 — The five-rung escalation ladder with persistence timers.

6.1 The HA side — Demand Follower blueprint

One blueprint (`config/blueprints/automation/dsc-hub/demand_follower.yaml`) drives all five appliances; the five automations are just bindings of demand entity → plug entity. Behaviour, defined once:

Event	Result	Why
Demand ON / OFF	Plug ON / OFF	Plain mirroring
Hub <code>unavailable</code> > 30 s	Plug forced OFF	A dead ESP must never leave a 750 W heater latched on; 30 s grace rides out OTA/reboot blips
HA restart	Plug re-synced to current demand	No stale states after HA maintenance
ESP reboot	Demands restore OFF → plugs OFF	By design; the ladder re-demands within its normal windows if the condition is still real

7 · OLED Interface — Menu Map & Navigation Logic

7.1 Physical controls

Control	Action
DIAL rotate	On Home: cycles the status carousel. In menus: moves cursor. On adjust pages: changes the value live (applied as you scroll). Detent acceleration: slow = 5%/detent, fast spin = 15%/detent.
DIAL push	Select / confirm ("push = done" on adjust pages).
BAK (GPIO14)	Back one level; cancels an adjust page. From Home: forces sleep. Hold ≈1 s = jump Home from anywhere.
CON (GPIO33)	Quick toggle of Tent Manual Override (fans) only — a deliberate panic hold. It cannot enter Full Auto: re-arming the curve is menu-only by design, so a stray press can never hand control back to automation.

7.2 Navigation map

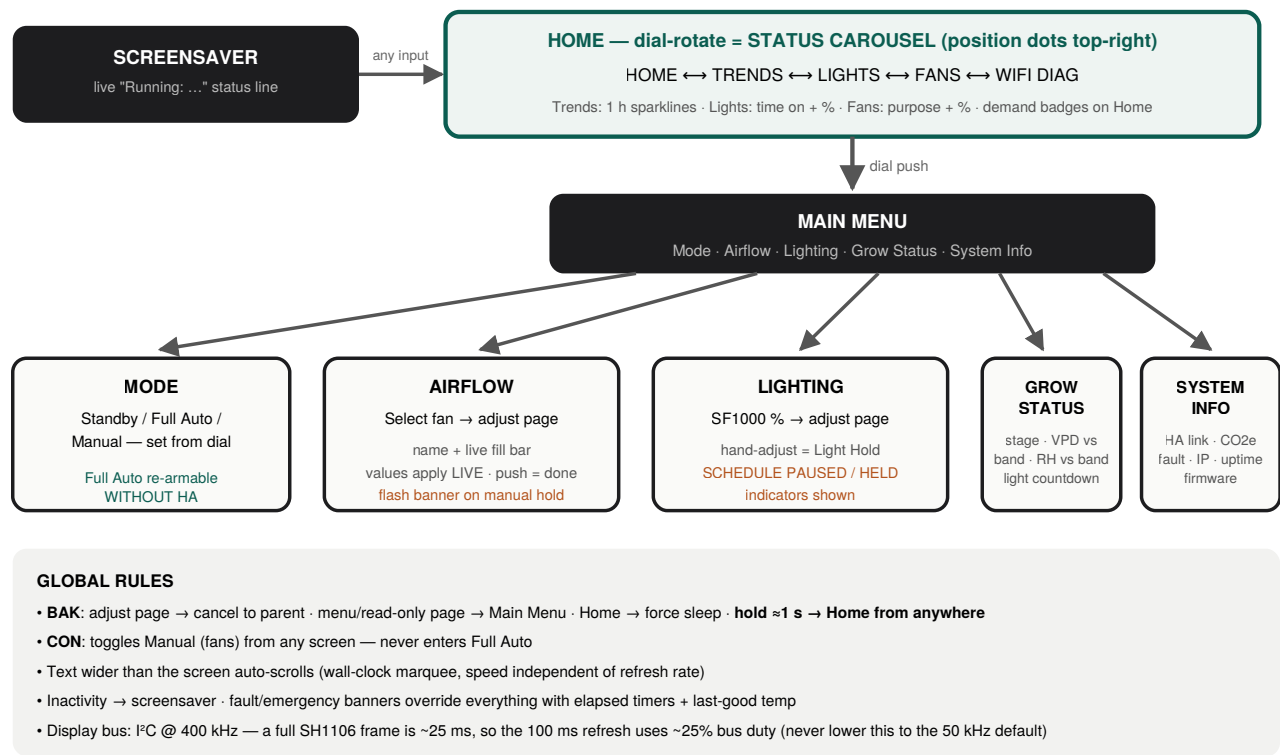


Fig. 6 — Menu logic map. Green = the Home carousel layer; black = menu layer; grey = global behaviours.

8 · Feature Reference

8.1 Climate & control

- **Ten-stage grow engine** (Germination → Dry Mode) with per-stage VPD / RH / temperature targets and photoperiod length; stage change loads defaults, fine-tuning persists across reboots.
- **Three control strategies**: VPD (default), Temperature, Humidity.
- **Two-axis software-defined damper**: separate OUT (heat) and ROOM (moisture) exhaust curves replacing a single fan curve; intakes slaved to total exhaust for guaranteed negative pressure; 15% fresh-air floor on OUT.
- **Five-rung appliance escalation ladder** with persistence timers and hysteresis (dehumidifier, humidifier, grow mat, 750 W heater, AC).
- **Device-local photoperiod** via SNTP (Australia/Sydney) with HA time backup; HH:MM Lights-On datetime; sunrise/sunset ramping with auto-scaling.
- **Split manual semantics**: Manual Override (fans) + self-healing Manual Light Hold (light).

8.2 Safety

- **Emergency failsafe** >35 °C: once-through purge, light cut, heat demands off, AC on; pre-trip state snapshot and restored on clear.
- **Sensor watchdog**: 3 min without a valid temperature → safe mode, all appliance demands off.
- **Safe-boot ventilation baseline**: any boot that isn't resuming Auto restores persisted manual fan targets, or defaults **Out 25% / Room 20% / Intakes 15%/15%** — a power blip can never strand a sealed tent with dead air.
- **Boot-resume prompt**: after power restoration mid-schedule, the OLED shows remaining window time with a 60 s auto-continue, then resumes automations from the correct photoperiod position.
- **HA-side fail-safe**: appliances forced off if the hub is unavailable >30 s.
- **Flash wear protection**: preference writes batched at 60 s so dial-spinning doesn't wear the chip.

8.3 UI

- Home carousel with position dots (Home / Trends / Lights / Fans / WiFi Diag); 1 h sparklines on Trends from the sensor ring buffer.
- Live-apply adjust pages with name + fill bar; 2 s inverted flash banner when a hold engages; SCHEDULE PAUSED / HELD indicators; demand badges on Home.
- Detent acceleration, marquee overflow scrolling, screensaver with live "Running: ..." automation narration, rich fault/emergency banners with elapsed timers and last-good temp.

9 · Home Assistant Integration

9.1 Entities exposed by DSC-HUB

Type	Entities
select	Grow Stage · Control Strategy
switch	Tent Full Auto Mode · Tent Manual Override (fans) · Manual Light Hold · Auto Photoperiod · Humidifier Auto/Demand · Dehumidifier Auto/Demand · Heater Auto/Demand · Grow Mat Auto/Demand · AC Auto/Demand
number	Target Temp · VPD Min · VPD Max · RH Min · RH Max · Sunrise Duration · Sunset Duration · SF1000 Target %
datetime	Lights-On Time (HH:MM, shows as AM/PM in HA)
binary_sensor	HA Link Status · Climate Sensor Fault
sensor	Tent Temperature · Tent Humidity · VPD · CO2e · Uptime · WiFi RSSI
text_sensor	IP Address · WiFi SSID · Firmware Version
button	Restart
fan	6 Inch Exhaust (Outside) · 6 Inch Exhaust (Room) · 4 Inch Intake (Main) · 4 Inch Intake (2×4)
light	SF1000

9.2 Division of labour

DSC-HUB owns	Home Assistant owns
Fan curves, damper blend, negative pressure Photoperiod timing + ramps Escalation ladder decisions (when to demand) All safety layers Local UI	Actually switching mains appliances (smart plugs) Demand Follower blueprint + 5 bindings Fail-safe cut on hub unavailability Dashboards, history, notifications Setting parameters (stage, targets, lights-on time)

9.3 Verification before trusting the ladder

1. Confirm demand entity IDs in **Developer Tools** → **States**, filter "demand" (ESPHome derives them from device + switch name, e.g. `switch.dsc_hub_heater_demand`).
2. Point each `switch.REPLACE_*` input in `dsc_hub_demand_automations.yaml` at the real plug entity (lists of multiple plugs are fine).
3. Force-test each rung once: toggle the demand switch manually and confirm the plug follows, then confirm the plug drops when you make the hub unavailable >30 s.

10 · Safety Systems & Boot Behaviour

Scenario	What happens
Power restored, Auto was on	Boot-resume prompt: shows remaining photoperiod window, 60 s auto-continue, resumes from the correct schedule position.
Power restored, Manual/ Standby was active	Safe-boot branch: persisted manual fan targets restored, or ventilation baseline 25/20/15/15. No dead-air state is possible.
WiFi / HA down	Photoperiod continues on SNTP-cached local time; fan curve continues; ladder keeps deciding (but plugs can't switch until HA returns — see FAQ).
DHT22 stops reporting (3 min)	Watchdog safe mode: safe ventilation, all appliance demands off, OLED fault banner with elapsed time and last-good temperature.
>35 °C	Failsafe: OUT 100 / ROOM 0 / intakes 100, light cut, heat off, AC demanded. State snapshot restored on clear; ladder timers reset.
ESP reboot (any cause)	Demand switches restore OFF → HA drops all plugs; Manual Light Hold cleared (schedule authoritative); stage re-applied at +3 s so targets are populated before control loops run.

11 · FAQ

Q — I just flashed rev 3 and the fans are sitting at 25 / 20 / 15 / 15. Is something broken?

No — that's the safe-boot ventilation baseline working. First power-up after flashing lands in the safe-boot branch if Auto wasn't armed. Set a mode (Full Auto from the Mode page or HA) and the curve takes over.

Q — I dimmed the SF1000 during the day and now the schedule "isn't working".

Touching the light engages **Manual Light Hold** — the schedule keeps ticking but stops driving the light, and the Lighting page shows HELD. It self-heals at the next lights-off boundary, so a daytime tweak resumes the schedule overnight. To resume immediately: toggle Auto Photoperiod, or turn Manual Override off.

Q — RH is over the max but the dehumidifier hasn't started. Why?

By design. The ladder gives the fans 5 minutes first, and only demands if RH is still above max + 3%. If the ROOM exhaust can move the moisture to the lung in time, the appliance never runs. If you want faster response for a specific situation, that's a manual toggle of the demand switch in HA, not a firmware change.

Q — Why can't the CON button put it back into Full Auto?

Deliberate asymmetry. CON is a panic hold — one press freezes the fans at their current values. Handing control *back* to the automation is consequential, so it's menu-only (Mode page) — a pocket press or a knock can never re-arm the curve.

Q — What actually happens if Home Assistant dies mid-flower?

Lights and fans: nothing changes — photoperiod and curves are device-local. Appliances: the ladder still *decides*, but the plugs can't follow until HA returns; meanwhile the HA-side fail-safe has already dropped them to OFF. The system degrades to "fans-only mode", which is the safe direction — it can under-heat or under-dry, never cook or ignite.

Q — Why is the OUT exhaust never below 15%?

CO₂. A fully closed recirculation loop starves the plants; the 15% floor continuously trades a little heat for fresh make-up air through the door undercut. If winter heating costs bite, the ladder's grow mat/heater rungs are the intended compensation, not lowering the floor.

Q — Summer heat is brutal. What's the play?

Shift Lights-On Time so the photoperiod runs overnight — lights-on coincides with cool ambient, the OUT exhaust does the rest, and the AC rung becomes a backstop instead of a daily runner. This is exactly why lights-on is an exposed HH:MM setting.

Q — The screen says SENSOR FAULT and everything appliance-related turned off.

The watchdog hasn't seen a valid temperature for 3 minutes. Check the DHT22 wiring/connector first (they're the most failure-prone part of the build). Ventilation continues at safe levels; appliances stay off because running heat into an unmeasured room is the one gamble the system refuses.

Q — Can I run the display bus slower / did the refresh change?

Leave I²C at 400 kHz. At ESPHome's 50 kHz default a full SH1106 frame takes ~200 ms of bus time against a 100 ms refresh — the loop becomes display-bound (laggy encoder, API stalls). This was a real, diagnosed fault class, not a preference.

Q — After a power cycle my appliances didn't come back on even though it's cold.

Correct sequence: demands restore OFF on reboot by design, HA drops the plugs, then the ladder re-demands within its normal persistence windows (1–5 min) if the condition is still real. Cold-start patience is the price of never latching a 750 W heater on unattended.

Q — Drying: should I turn the fans up to speed it along?

No. Dry Mode sets ~19 °C / ~60% RH / dark / low fans, and drying genuinely wants low, indirect airflow over 7–14 days. The ~20% exhaust baseline is the feature; blasting it is how you get hay.

12 · Notes, Known Issues & Outstanding Work

12.1 Hardware cautions

The latch-up incident (June 2026) — do not repeat

An ESP32 WROOM-32 was destroyed by overvoltage from a PWM driver module's gate-drive circuitry back-feeding a GPIO: suspected CMOS latch-up plus probable regulator failure (frozen screen, solid status LED, OTA dead, unrecoverable). **Before any replacement board goes into service:** 330 Ω–1 kΩ series resistor + 3.3 V zener or Schottky clamp on every PWM signal line, or an opto-isolated gate driver. This is a standing rule for all Digital-Plant ESP nodes driving external power modules.

12.2 Operational notes

- **Compile locally before flashing.** The rev 3 config is schema-validated and codegen-verified, but `esphome run` on your machine is the real gate (the sandbox can't run the full C++ build). The display lambda changed most in rev 3 — that's where any compile complaint would surface.
- **OTA fallback:** the device broadcasts `DSC-HUB Fallback Hotspot` if WiFi credentials fail.
- **Timezone is critical:** the SNTP platform is pinned to `Australia/Sydney` ; without it the photoperiod runs on UTC.
- **Firmware source of record:** `/mnt/user-data/outputs/dsc-hub.yaml` (rev 3, ~1,631 lines) from the July 2026 QA session; HA-side files are `demand_follower.yaml` (blueprint) + `dsc_hub_demand_automations.yaml` (5 bindings).

12.3 Outstanding work

Item	Detail	Status
HA dashboard migration	Entity renames: Main/Aux → Outside/Room fans; <code>temp / humidity</code> → <code>tent_temperature / tent_humidity</code> ; Lights-On Hour → <code>datetime</code> ; 250 W light removed; four dial/button entities removed (delete references). New entities to place: Manual Light Hold, Uptime, Firmware Version, Restart.	Pending
Demand automation bindings	Replace the five <code>switch.REPLACE_*</code> placeholders with real plug entity IDs; run the §9.3 verification.	Pending verification
Notion archive	File this manual + firmware revision notes under Digital-Plant so the six-month rule holds.	Suggested

12.4 Revision history

Rev	Date	Summary
1	Jun 2026	Full rewrite from legacy config: ten-stage engine, VPD/Temp/RH strategies, SNTP photoperiod, humidifier/dehumidifier demands, watchdog, failsafe snapshot/restore, Grow Status page. Fixed: unreachable Full Auto, fans stuck at 100% post-failsafe, heat emergency not cutting lights, dead wake-guard, hardcoded HA link status.
2	Early Jul 2026	~1094 → ~1631 lines. 250 W light removed → 4th fan (GPIO19); software-defined damper (Outside/Room split, two-axis controller); HH:MM lights-on datetime; sunrise/sunset ramps; five-rung escalation ladder with persistence timers; live-apply adjust pages; marquee; screensaver status line; home carousel; boot-resume prompt; OLED regrid (16/27/38/49 rows for Roboto-10 descenders).
3	7 Jul 2026	QA pass (3 critical, 4 high, 6 medium fixed). Headliners: safe-boot ventilation baseline + persisted manual targets (C1 dead-air fix); I ² C 400 kHz (C2 display-bound loop fix); split manual semantics with self-healing Manual Light Hold; ladder timers reset on failsafe/watchdog; sparklines, detent acceleration, hold-Back-to-Home, richer banners; Uptime/Firmware/Restart entities; dial+buttons made internal. HA blueprint + 5 demand bindings delivered.